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Prop fn holomorfa iff wirtinger

Proposición 1. *Sea $f \in \mathcal{C}^1(\Omega)$ con $\Omega \subset \mathbb{C}$ abierto, entonces*

(1) *f es holomorfa en $\Omega \iff \forall z \in \Omega : \dfrac{\partial f}{\partial \overline{z}}(z) = 0.$*

(2) *f es antiholomorfa en $\Omega \iff \forall z \in \Omega : \dfrac{\partial f}{\partial z}(z) = 0.$*

***Demostración*:** En ambos casos, procedemos por definición de las derivadas de Wirtinger:

(1) Escribimos $z = x + iy$ con $x, y \in \mathbb{R}$ y $f = u + iv$ con $u = \Re(f), v = \Im(f)$. Entonces,

$$\forall z \in \Omega : \frac{\partial f}{\partial \overline{z}}(z) = 0 \iff \frac{1}{2}\left(\frac{\partial f}{\partial x}(z) + i\,\frac{\partial f}{\partial y}(z)\right) = 0 \iff \frac{\partial f}{\partial x}(z) + i\,\frac{\partial f}{\partial y}(z) = 0$$
$$\iff \frac{\partial u}{\partial x}(z) + i\,\frac{\partial v}{\partial x}(z) + i\,\frac{\partial u}{\partial y}(z) - \frac{\partial v}{\partial y}(z) = 0.$$

Tomando ahora partes real e imaginaria tenemos que esto es equivalente a

$$\frac{\partial u}{\partial x}(z) - \frac{\partial v}{\partial y}(z) = 0 \quad \wedge \quad \frac{\partial v}{\partial x}(z) + \frac{\partial u}{\partial y}(z) = 0,$$

que son las Ecuaciones de Cauchy-Riemann, luego esto es equivalente a que f sea holomorfa en Ω. Es decir, f es holomorfa en $\Omega \iff \forall z \in \Omega : \frac{\partial f}{\partial \overline{z}}(z) = 0.$

(2) De forma análoga, tenemos que

$$\forall z \in \Omega : \frac{\partial f}{\partial z}(z) = 0 \iff \frac{\partial u}{\partial x}(z) + i\,\frac{\partial v}{\partial x}(z) - i\,\frac{\partial u}{\partial y}(z) + \frac{\partial v}{\partial y}(z) = 0$$
$$\iff \frac{\partial u}{\partial x}(z) + \frac{\partial v}{\partial y}(z) = 0 \quad \wedge \quad \frac{\partial v}{\partial x}(z) - \frac{\partial u}{\partial y}(z) = 0.$$

Por otro lado, $\overline{f} = u - iv$ y, por tanto, $\frac{\partial \overline{f}}{\partial x} = \frac{\partial u}{\partial x} - i\,\frac{\partial v}{\partial x}$ y $\frac{\partial \overline{f}}{\partial y} = \frac{\partial u}{\partial y} - i\,\frac{\partial v}{\partial y}$. Por tanto, por el Teorema de Cauchy-Riemann, $\overline{f}$ es holomorfa en Ω

$$\iff \frac{\partial u}{\partial x}(z) + \frac{\partial v}{\partial y}(z) = 0 \quad \wedge \quad -\frac{\partial v}{\partial x}(z) + \frac{\partial u}{\partial y}(z) = 0,$$

que son precisamente las ecuaciones anteriores. Por tanto, f es antiholomorfa en $\Omega \iff \forall z \in \Omega : \frac{\partial f}{\partial z}(z) = 0.$

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Referenciado en

- `Lem-laplaciano-cambio-variable`

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